



SKG14 IC8 Pacing and Emerging Technologies for ICM Engineers, L2

21.0 - Blockchain

July 2025

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Learning Outcome

At the end of the session, you will be able to:

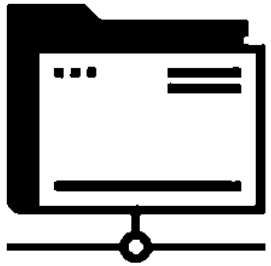
- Identify the core concepts, structure, and key features of blockchain technology, including decentralization, immutability, and consensus mechanisms.
- Explain relevant international standards (ISO, IEEE, NIST) and regulatory considerations such as data privacy, legal recognition, and compliance frameworks.
- Explore practical applications of blockchain in industries like finance, supply chain, and healthcare, and explain how smart contracts enhance efficiency and trust.

Pre-Quiz

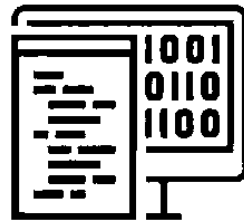
Interactive Learning

21.1 Introduction to Blockchain Technology

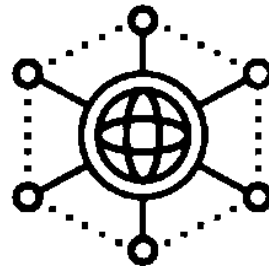
Introduction to Blockchain Technology



A digital ledger that keeps a record of all transactions taking place on a peer-to-peer network



All information transferred via blockchain is encrypted, and every occurrence recorded, meaning it cannot be altered



It is decentralized, so there's no need for any central, certifying authority



It can be used for much more than the transfer of currency; contracts, records and other kinds of data can be shared



Encrypted information can be shared across multiple providers without risk of a privacy breach

Source: <https://www.process.st/introduction-to-blockchain/>

Definition of Blockchain Technology

International Organization for Standardization (ISO)

A distributed ledger with confirmed blocks organized in an append-only, sequential chain using cryptographic links

Provides foundational terminology for blockchain and distributed ledger technologies.

The National Institute of Standards and Technology (NIST)

Collaborative, tamper-resistant ledger that maintains transactional records.

Each block in the chain contains a unique identifier linked to the previous block, ensuring the integrity and security of the data.

Institute of Electrical and Electronics Engineers (IEEE)

A disruptive technology for building consensus and trust in a peer-to-peer network without centralized control.

Emphasizes blockchain's ability to create a secure and decentralized ledger of transactions, which was first implemented in Bitcoin, the pioneering cryptocurrency.

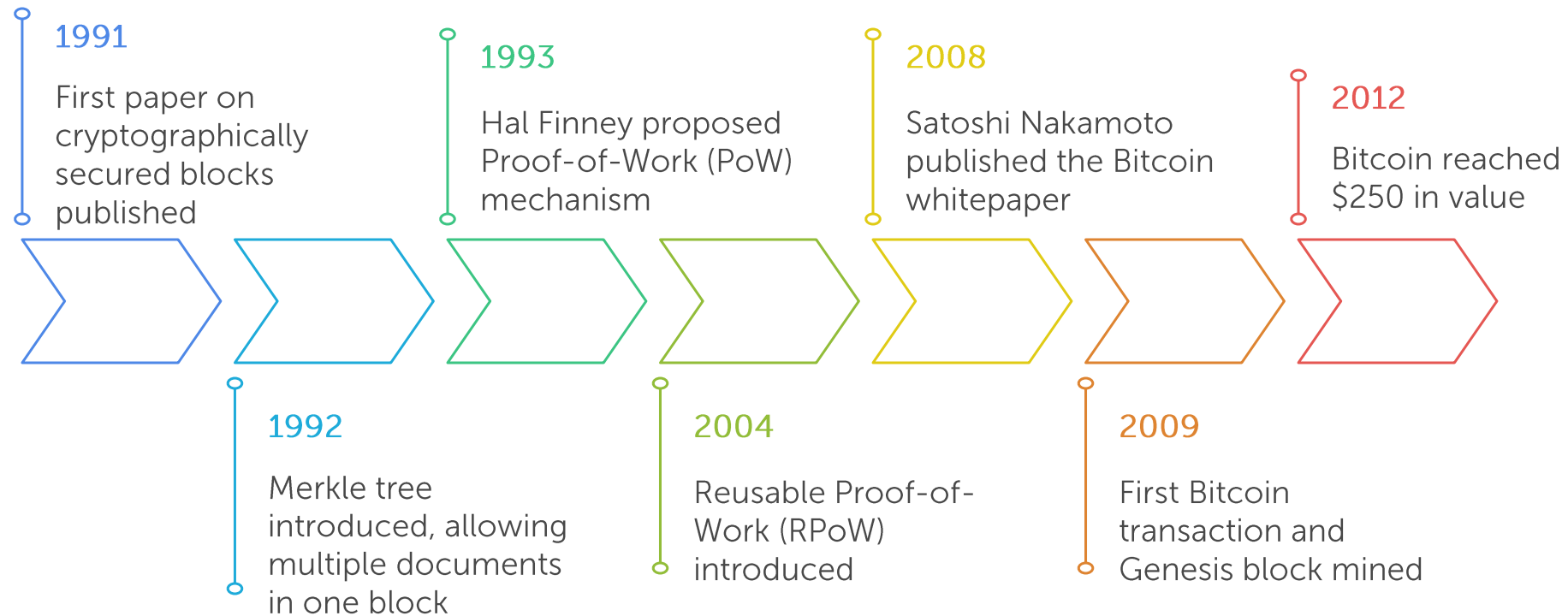
Definition of Blockchain Technology

Based on the three standards, the simplified BLOCKCHAIN as:

- ✓ A distributed, tamper-resistant ledger that organizes transactional data into sequential, cryptographically linked blocks in an append-only manner.
- ✓ It operates in a decentralized peer-to-peer network without centralized control, ensuring data integrity, security, and consensus.
- ✓ Initially popularized by Bitcoin, blockchain is recognized for its ability to build trust and transparency in various applications.

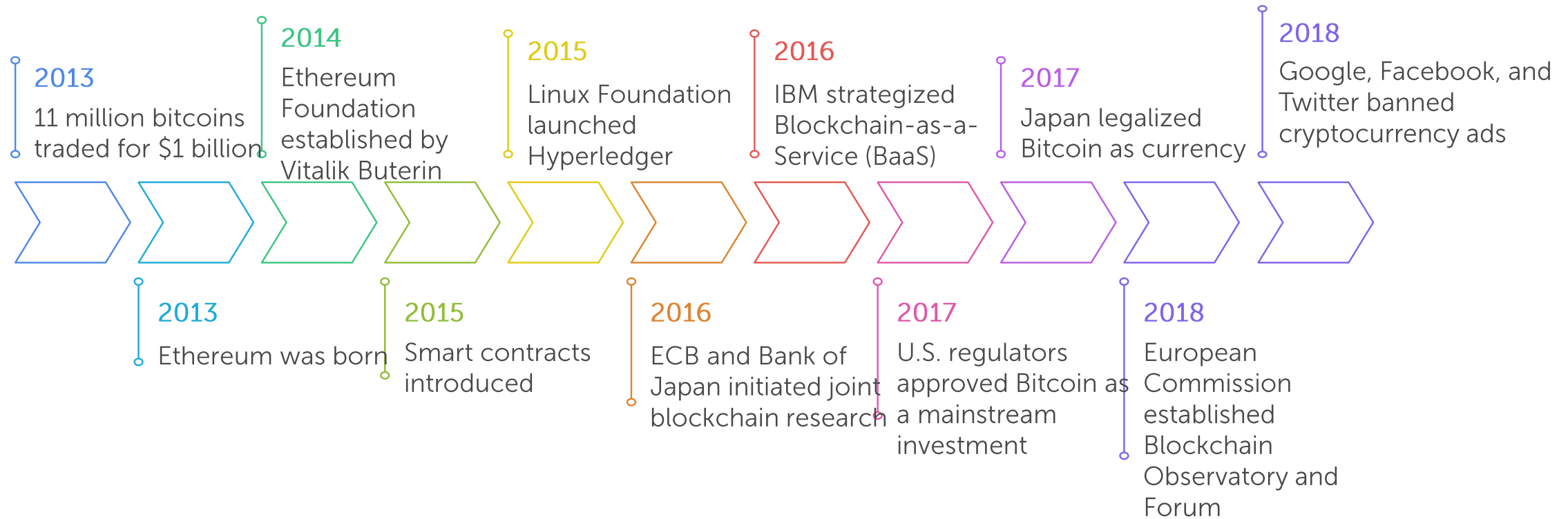


Historical Background and Evolution



Source: <https://www.shiksha.com/online-courses/articles/evolution-of-blockchain-technology/>

Historical Background and Evolution



Source: <https://www.shiksha.com/online-courses/articles/evolution-of-blockchain-technology/>

21.2 Standards, References & Regulations

Existing Standards and Frameworks



International Organization for Standardization (ISO)

- ISO has established Technical Committee 307 (ISO/TC 307) dedicated to blockchain and distributed ledger technologies.
- This committee addresses aspects such as terminology, reference architecture, security, privacy, and interoperability.
- Notably, ISO 22739:2020 provides foundational terminology for blockchain and distributed ledger technologies.



Institute of Electrical and Electronics Engineers (IEEE)

- IEEE has formed the Blockchain and Distributed Ledger Standards Committee, which oversees the development of standards in this domain.
- For instance, IEEE 2418.7-2021 defines a baseline architectural framework for the use of blockchain in supply chain finance.



National Institute of Standards and Technology (NIST)

- NIST provides comprehensive resources on blockchain technology, including high-level technical documents that break down key components, emerging standards, and system architectures supporting blockchain

Existing Standards and Frameworks



GS1

- GS1 offers global standards for identification and structured data, enabling blockchain network users to scale enterprise adoption.
- Their standards help maintain a single, shared version of the truth about supply chain and logistics events, increasing data integrity and trust between parties.



European Commission

- The European Commission actively engages with the blockchain standards community, working closely with relevant bodies worldwide to ensure interoperability and support the technology's development.

Regulatory Considerations

1. **Data Privacy:** Ensure compliance with data protection laws (e.g., GDPR) by encrypting or off-chaining sensitive information.
 - Use encryption and pseudonymization for personal data.
 - Implement off-chain storage for sensitive data.
2. **Legal Recognition:** Confirm blockchain records and smart contracts are legally enforceable in the applicable jurisdiction.
 - Ensure compliance with local electronic signature laws.
 - Draft smart contracts with clear legal terms.
3. **Anti-Money Laundering (AML) and Know Your Customer (KYC) Compliance:** Implement systems to prevent money laundering and verify user identities.
 - Implement robust KYC processes for onboarding users.
 - Use blockchain analytics tools to monitor and report suspicious transactions.

Regulatory Considerations

4. **Taxation:** Understand and comply with tax obligations for blockchain activities, maintaining transparent records.
 - Understand the tax implications of blockchain activities in relevant jurisdictions.
 - Maintain transparent and accurate records for tax reporting.
5. **Securities Regulation:** Classify and register tokens appropriately to comply with securities laws.
 - Classify tokens appropriately (utility vs. security tokens).
 - Comply with securities registration and reporting requirements where applicable
6. **Intellectual Property:** Protect proprietary technologies and address licensing and ownership of blockchain assets.
 - Protect proprietary blockchain solutions through patents or copyrights.
 - Address licensing and ownership rights for blockchain-based assets.

Regulatory Considerations

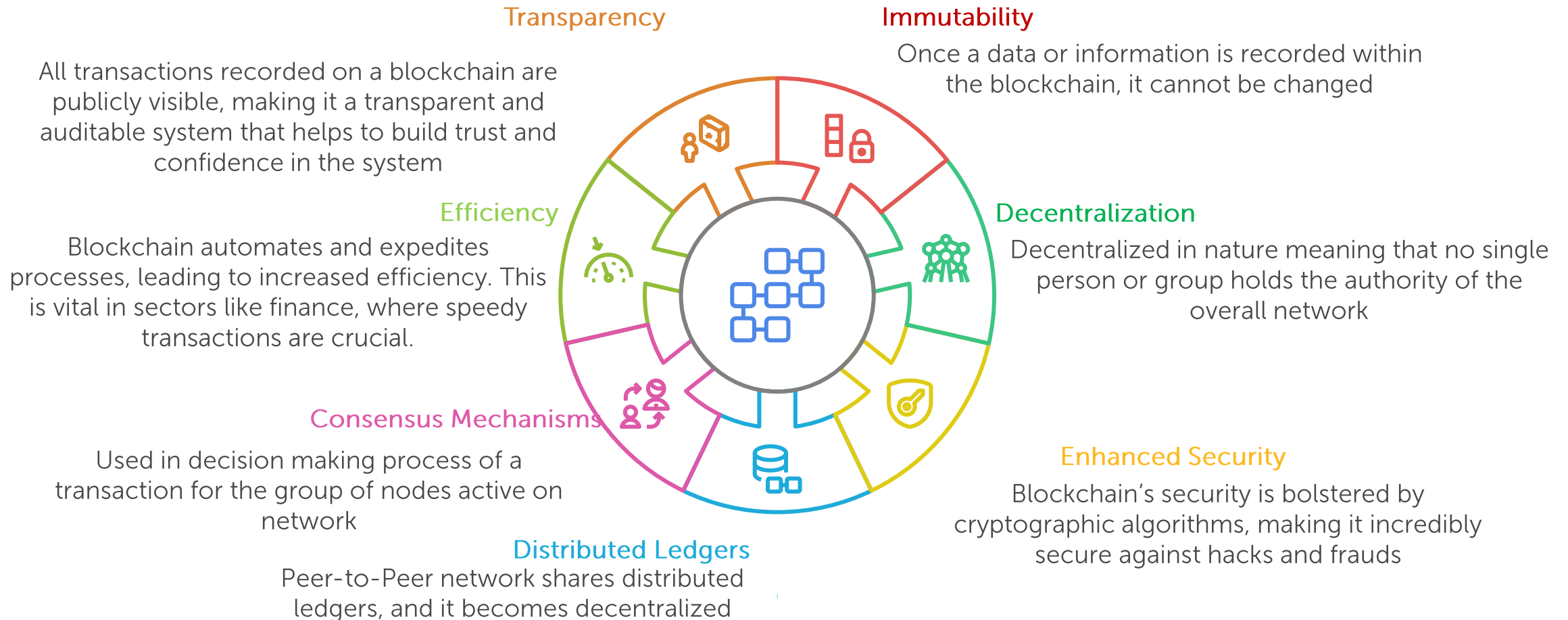
7. **Jurisdictional Issues:** Define applicable jurisdictions for governance and dispute resolution
 - Identify the applicable legal jurisdiction for disputes and enforcement.
 - Align blockchain governance with international legal frameworks
8. **Cybersecurity:** Adhere to security standards to safeguard blockchain networks and prevent breaches
 - Follow standards such as ISO/IEC 27001 for information security.
 - Conduct regular audits and risk assessments.
9. **Consumer Protection:** Provide transparency, clear terms of use, and dispute resolution mechanisms
 - Develop transparent terms of use and privacy policies.
 - Implement mechanisms for dispute resolution and user education.

Regulatory Considerations

- 
10. **Environmental Impact:** Use energy-efficient consensus mechanisms and monitor sustainability compliance.
 - Use energy-efficient alternatives (e.g., Proof of Stake).
 - Monitor and report energy consumption to comply with sustainability goals
 11. **Industry-Specific Rules:** Align with regulations for sectors like healthcare (e.g., HIPAA) and finance (e.g., PSD2)
 - Align with standards such as HIPAA (healthcare) or PSD2 (finance).
 - Tailor blockchain solutions to meet sector-specific compliance needs.
 12. **Government and Accountability:** Establish clear roles, responsibilities, and transparent governance frameworks.
 - Define roles and responsibilities for network participants
 - Ensure accountability through transparent governance frameworks.

21.3 Blockchain Fundamentals

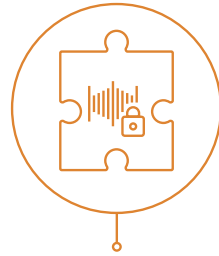
Features of Blockchain



Immutability

Data Integrity

Immutability ensures data remains accurate and trustworthy. Ideal for finance, healthcare and legal.

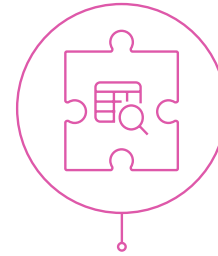


Fraud Prevention

Inability to alter recorded data protects against unauthorized changes. Reduces fraud risks and tampering.

Transparency

Immutable records foster trust among participants. All transactions are publicly verifiable.

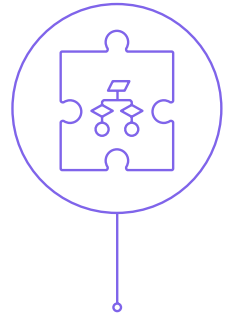
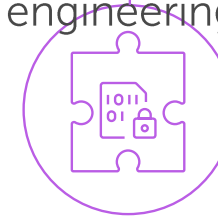


Audit Simplification

Immutable data simplifies auditing processes. Records can be easily verified against blockchain transactions.

Cryptographic Hashing

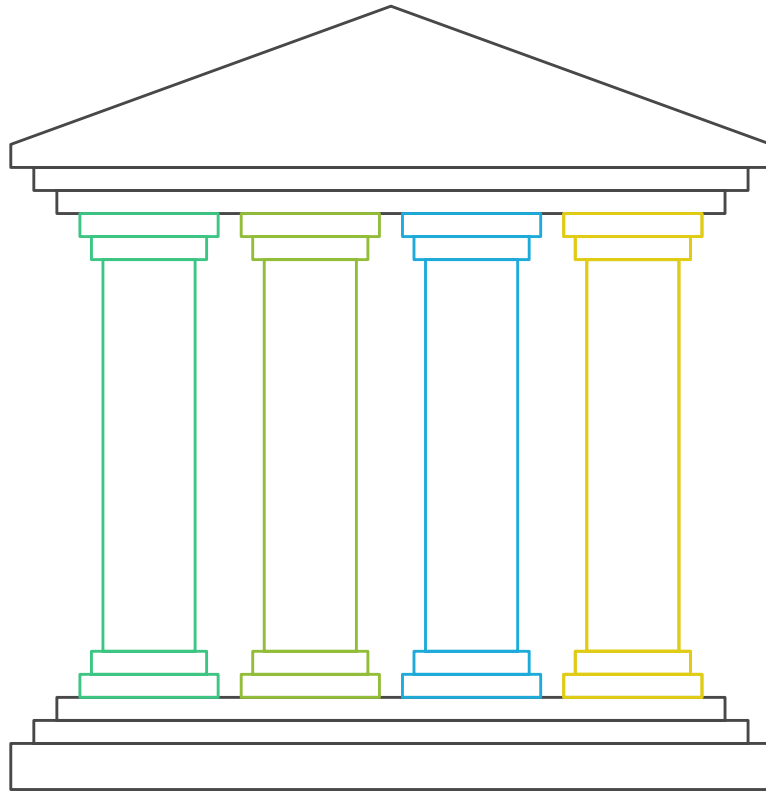
Blockchain achieves immutability through cryptographic hashes. Uniquely identifies each block and prevents reverse engineering.



Merkle Tree

This data structure ensures that transactions remain unchanged across the network.

Decentralization



Trustless Environment

Eliminates the need for trust among participants through network verification.



Resilience

Enhances system robustness by reducing single points of failure.



Transparency

Fosters openness and accountability by providing equal data access.



Equal Participation

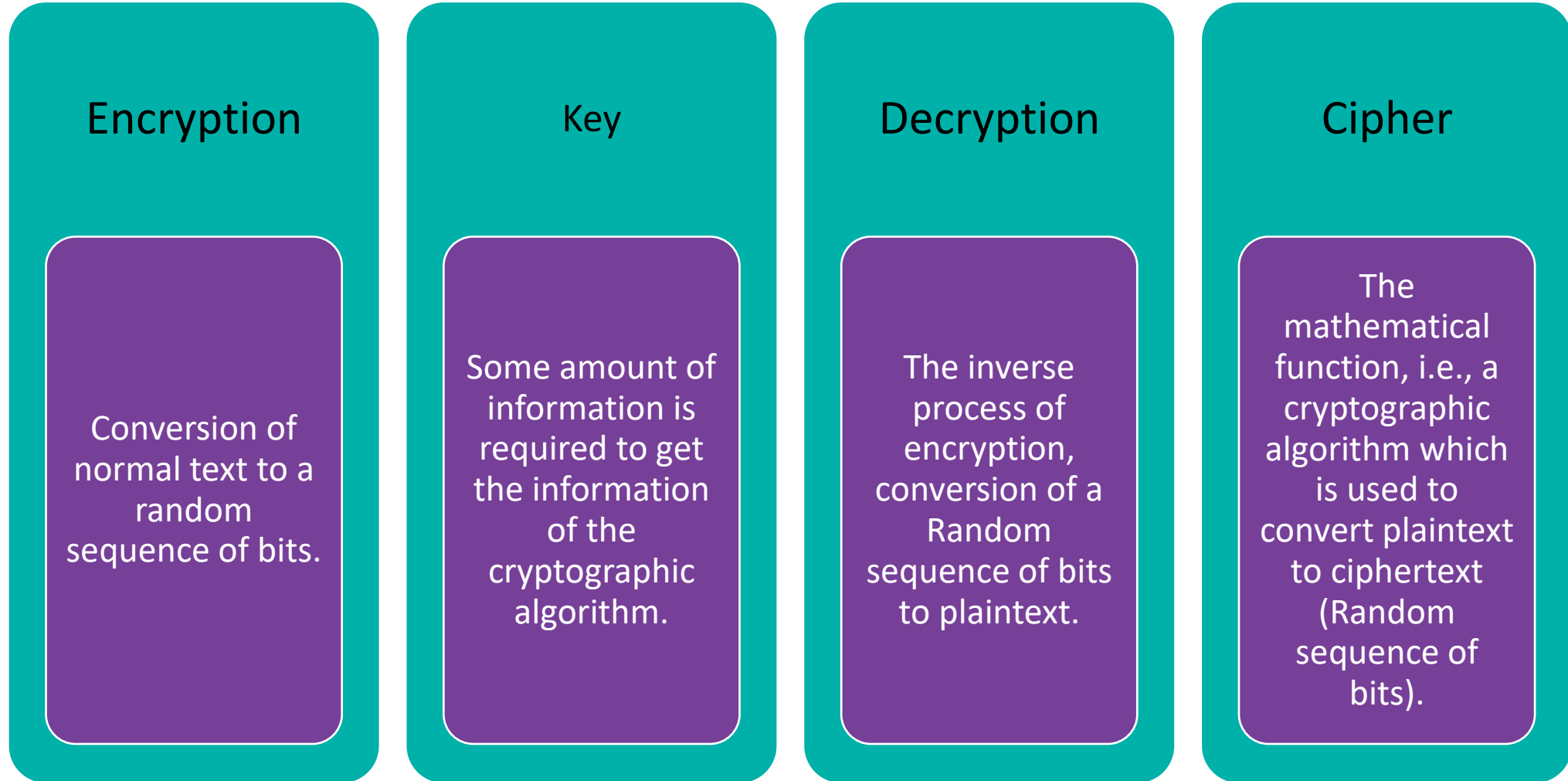
Ensures all nodes have equal authority, preventing monopolization.

Enhanced Security

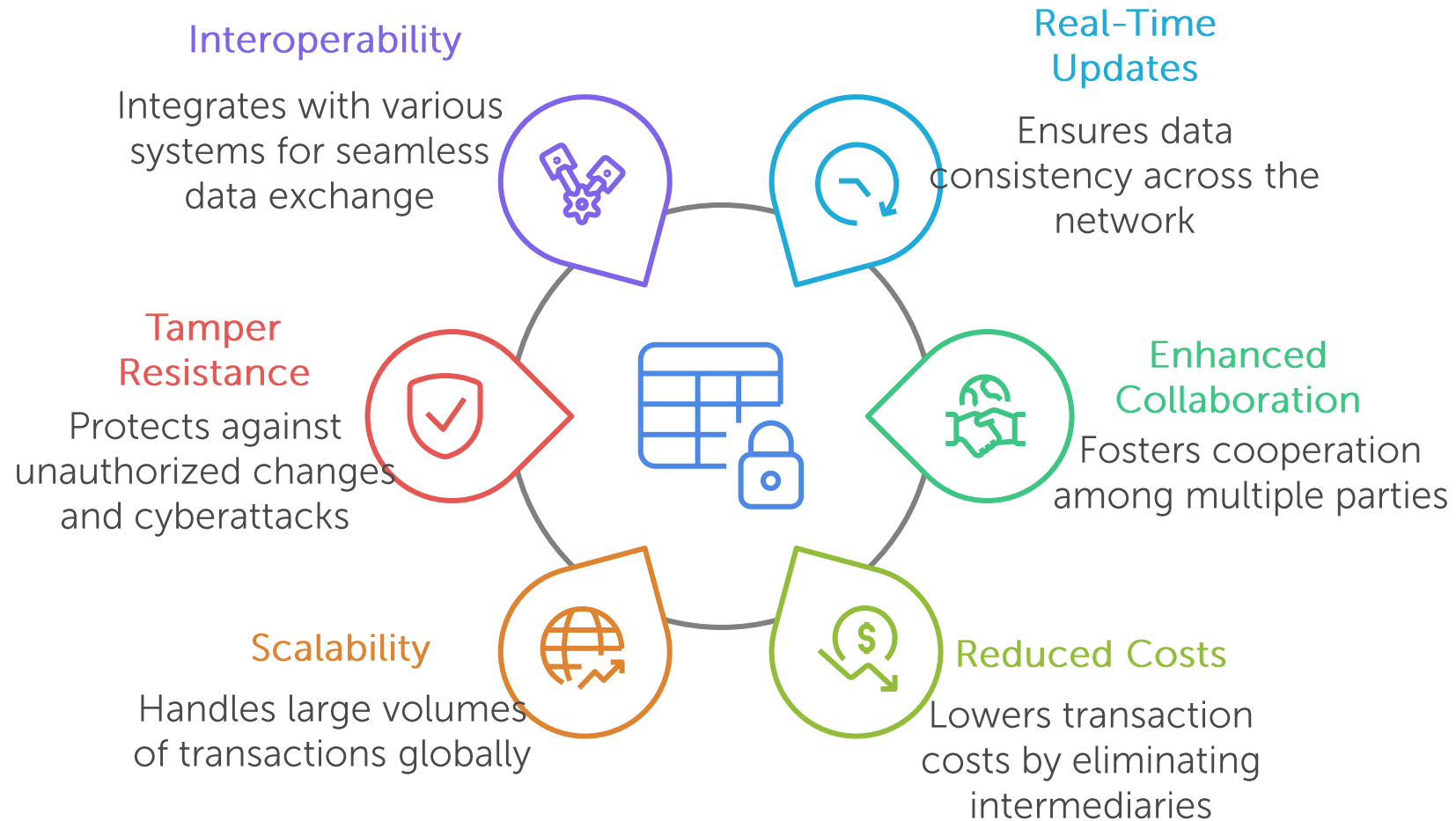
- ✓ Blockchain security is the risk management procedure or system that is designed to protect a blockchain network from online threat actors and cybercriminals.
- ✓ Blockchain security uses a combination of cybersecurity best practices, tested frameworks and technical assurances to protect against fraud and cyberattacks.
- ✓ Even though blockchains are cryptographically signed, blockchain security is needed to ensure that only authorized users gain access to nodes and the blockchain network overall.
- ✓ Blockchain networks take advantage of several cybersecurity protocols and frameworks to ensure sensitive user information is kept secure and hidden.

Enhanced Security

Cryptography and Encryption

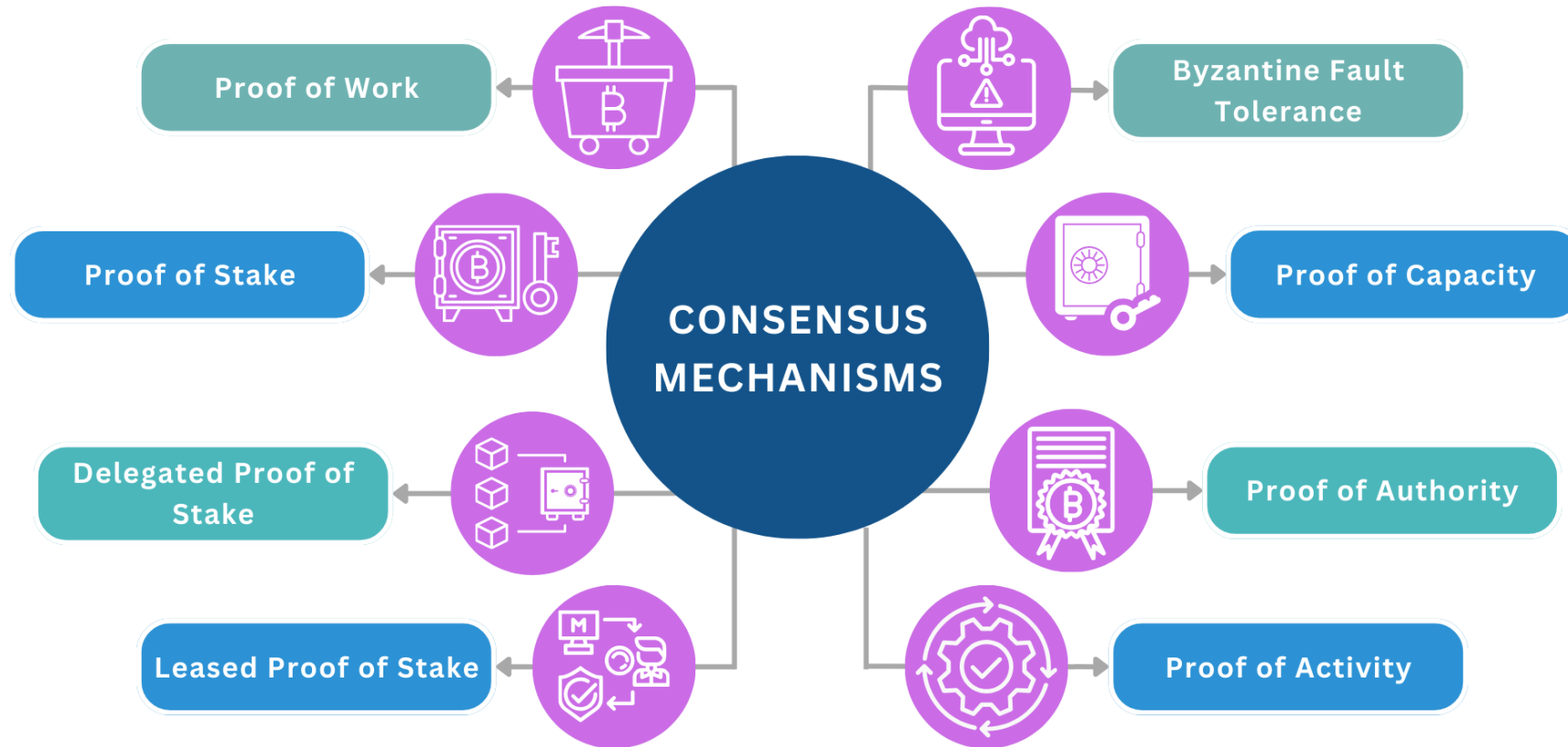


Distributed Ledgers



Consensus Mechanisms

Consensus mechanisms are at the heart of blockchain technologies, enabling distributed networks to agree on the state of the ledger in a trustless environment.

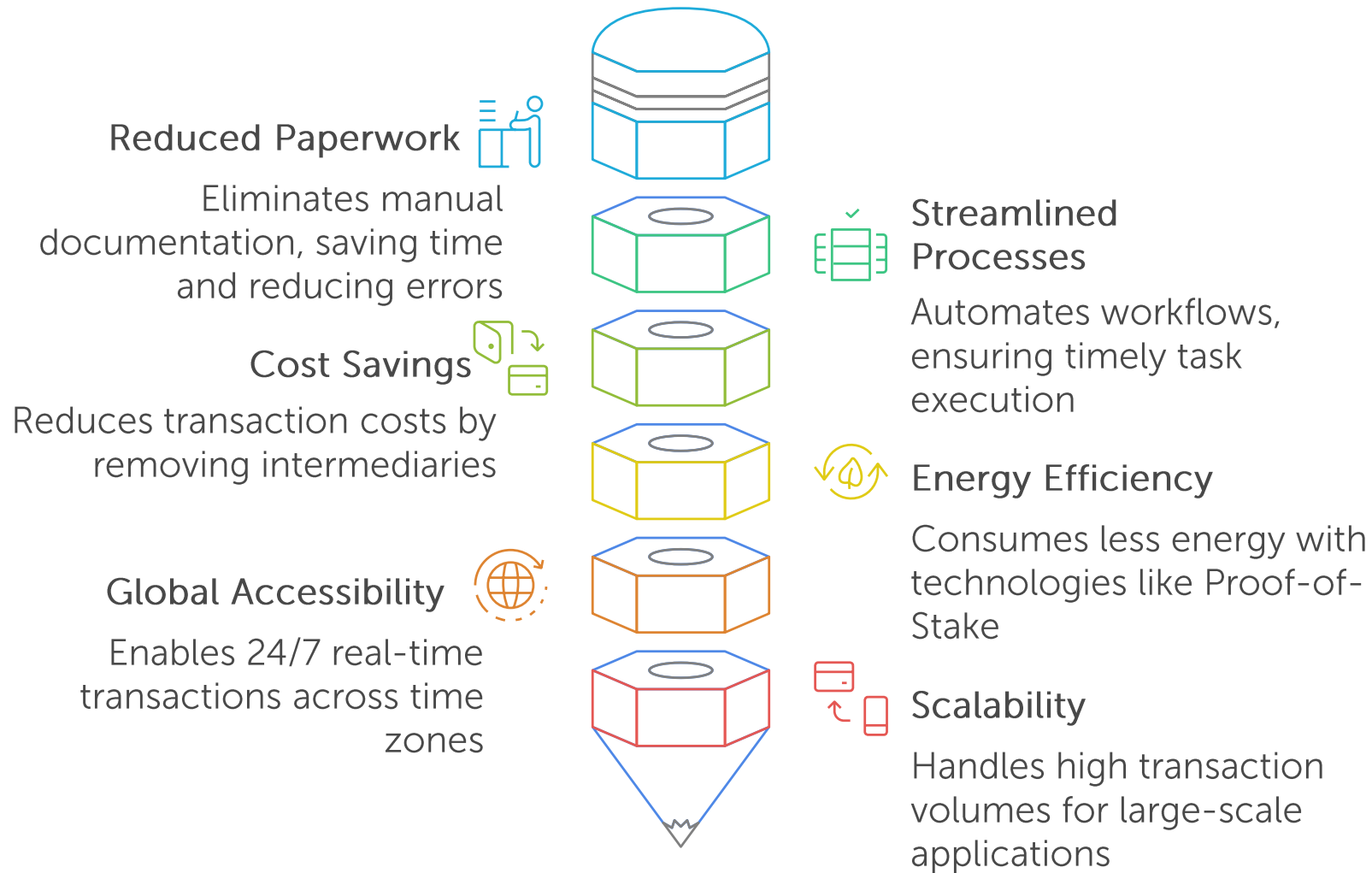


Consensus Mechanisms

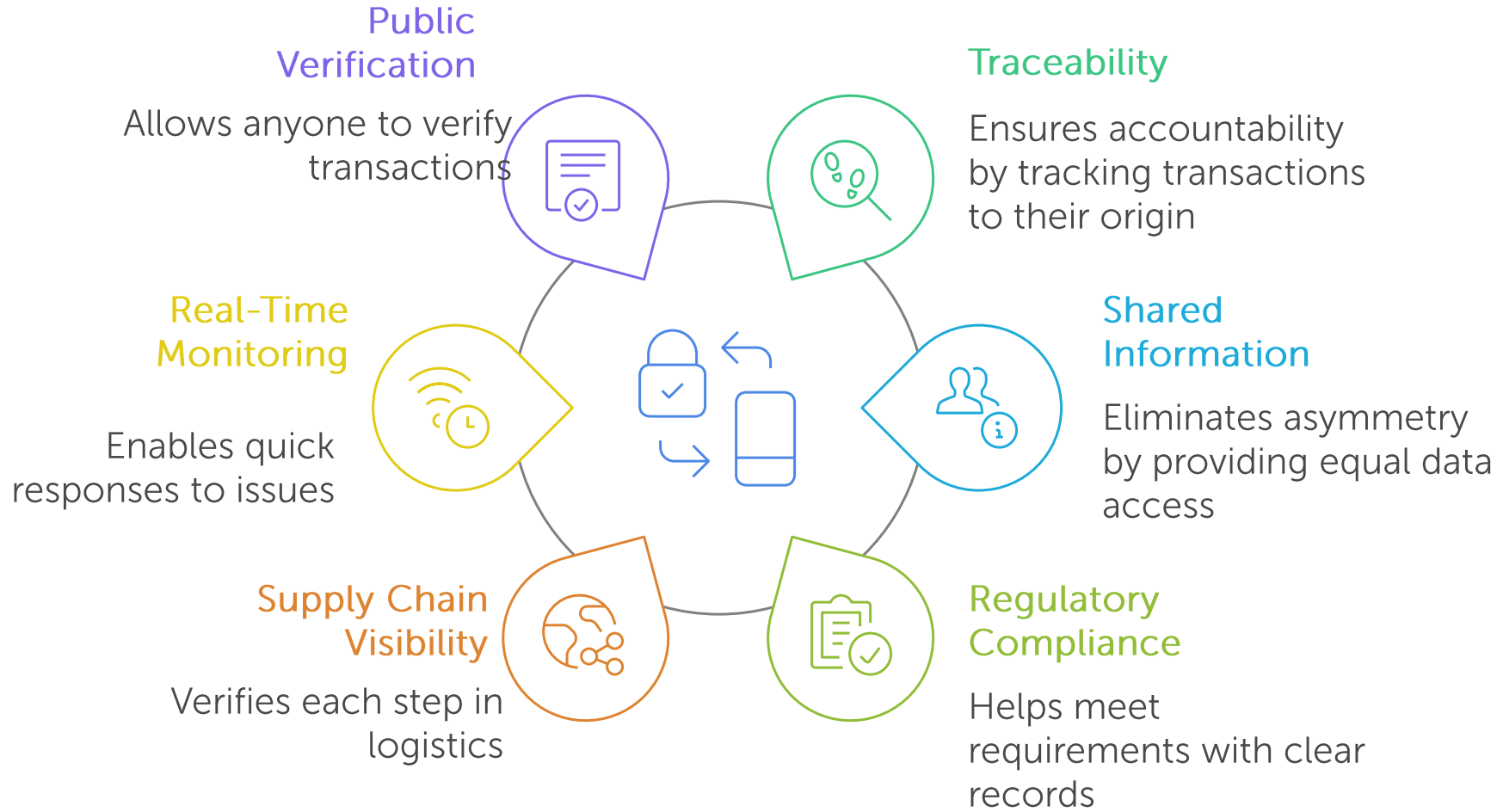
Mechanism	Description
Proof of Work (PoW)	A consensus mechanism where participants, called miners, compete to solve complex mathematical puzzles using computational power. The first to solve the puzzle gets to validate transactions and add a new block to the blockchain. While secure and widely used (e.g., in Bitcoin), PoW is energy-intensive and requires significant hardware resources.
Proof of Stake (PoS)	A consensus mechanism where validators are chosen based on the amount of cryptocurrency they "stake" or lock up as collateral. The higher the stake, the greater the chance of being selected to validate transactions and create new blocks. PoS is more energy-efficient than PoW and promotes holding over spending.
Delegated Proof of Stake (DPoS)	An evolution of PoS where token holders vote for a small number of delegates who are responsible for validating transactions and producing blocks. DPoS increases transaction speed and efficiency while still incorporating a degree of decentralization through stakeholder voting.
Leased Proof of Stake (LPoS)	A variation of PoS where users can lease their staked coins to a full node or validator. This leased stake increases the chances of the validator being selected to create the next block, and the rewards are then shared with those who leased their tokens. It encourages broader participation without running a full node.
Byzantine Fault Tolerance (BFT)	A consensus model designed to function correctly even if some participants (nodes) act maliciously or fail. BFT-based mechanisms ensure that all honest nodes agree on the same state of the blockchain despite faulty or compromised nodes. Often used in systems where fault tolerance and security are critical.
Proof of Capacity (PoC)	Allows network participants to use their available storage space on hard drives to participate in the mining process. Participants pre-generate solutions and store them on disk, which are later used to validate transactions and create new blocks. It offers a more energy-efficient alternative to Proof of Work.
Proof of Authority (PoA)	A consensus mechanism where transactions and blocks are validated by approved accounts, known as validators. These validators are pre-selected and trusted entities, making PoA suitable for permissioned blockchain networks where speed and efficiency are prioritized over decentralization.
Proof of Activity (PoA)	A hybrid consensus mechanism combining features of Proof of Work and Proof of Stake. Mining begins with PoW, but once a block is mined, a group of randomly selected validators (based on PoS) sign off on it. This blend improves security, decentralization, and energy efficiency compared to PoW alone.

Source: <https://mdpi.com/2076-3417/13/3/1353> © 2025 Petrolia Nasional Berhad (PETRONAS) | 33

Efficiency



Transparency



Key Components: Blocks, Transactions, and Nodes

Structure of Blocks

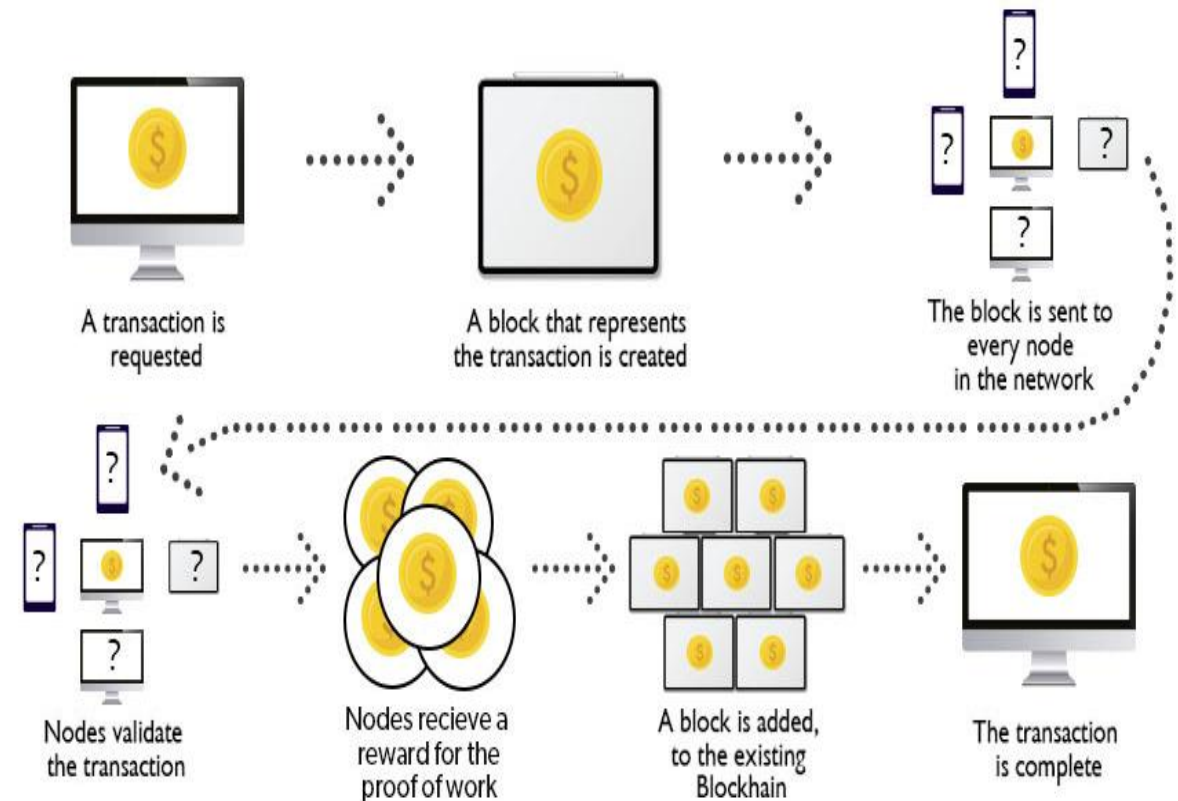
- ✓ Each block in a blockchain contains transaction data, a timestamp, and a hash of the previous block, ensuring secure data linking.

Role of Nodes

- ✓ Nodes are the participants in the blockchain network that validate transactions and maintain the integrity of the entire blockchain

Transaction Flow

- ✓ Transactions are grouped into blocks and added to the blockchain, providing a transparent and secure record of all activities.



Source: <https://doi.org/10.1007/s11042-023-15401-1>

How Blockchain Works



Network of Nodes

- Blockchain functions through a decentralized network of nodes that validate and approve transactions. This structure enhances security and transparency.

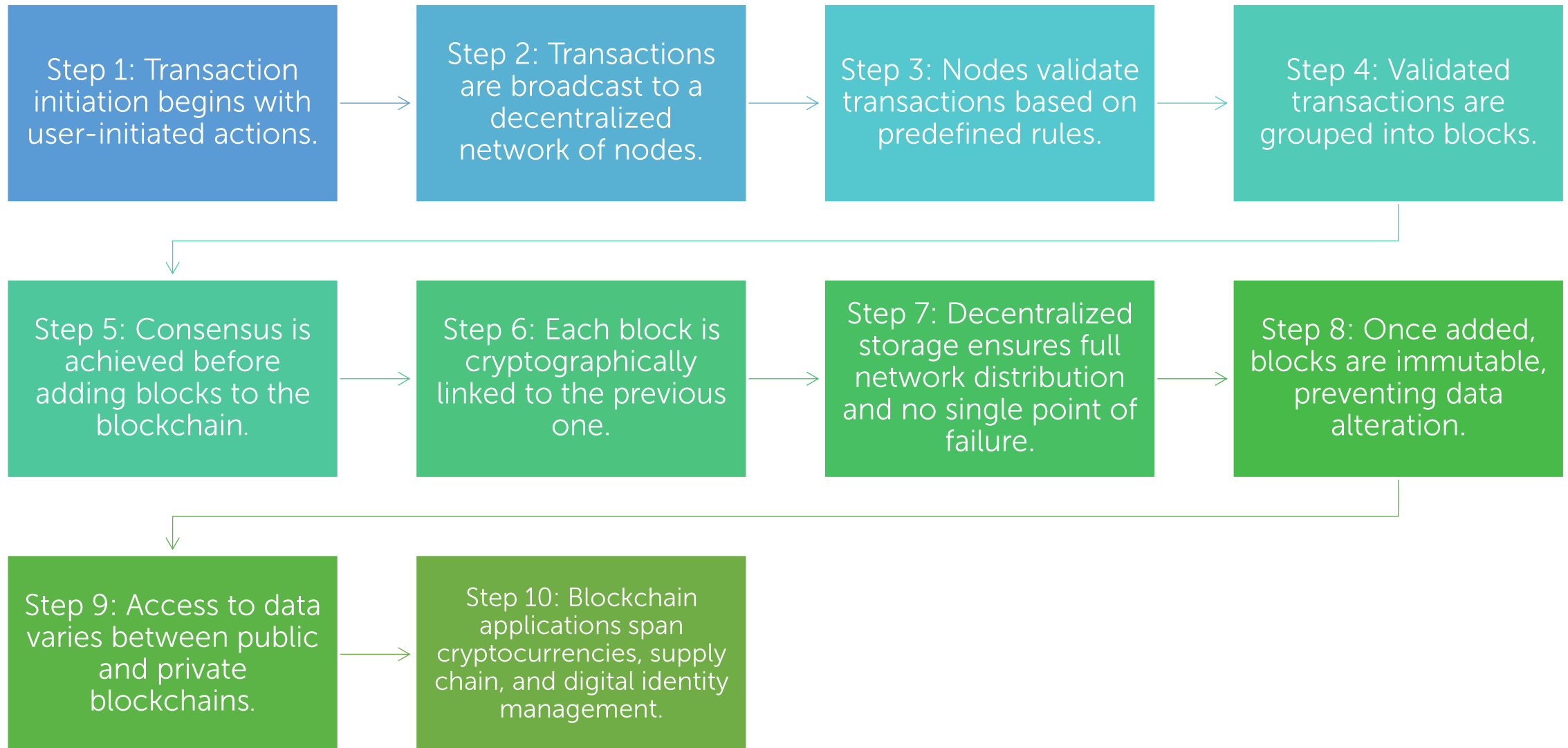
Blocks and Transactions

- Transactions are recorded in blocks, which are linked to create a chain. Each block contains transaction data and a reference to the previous block.

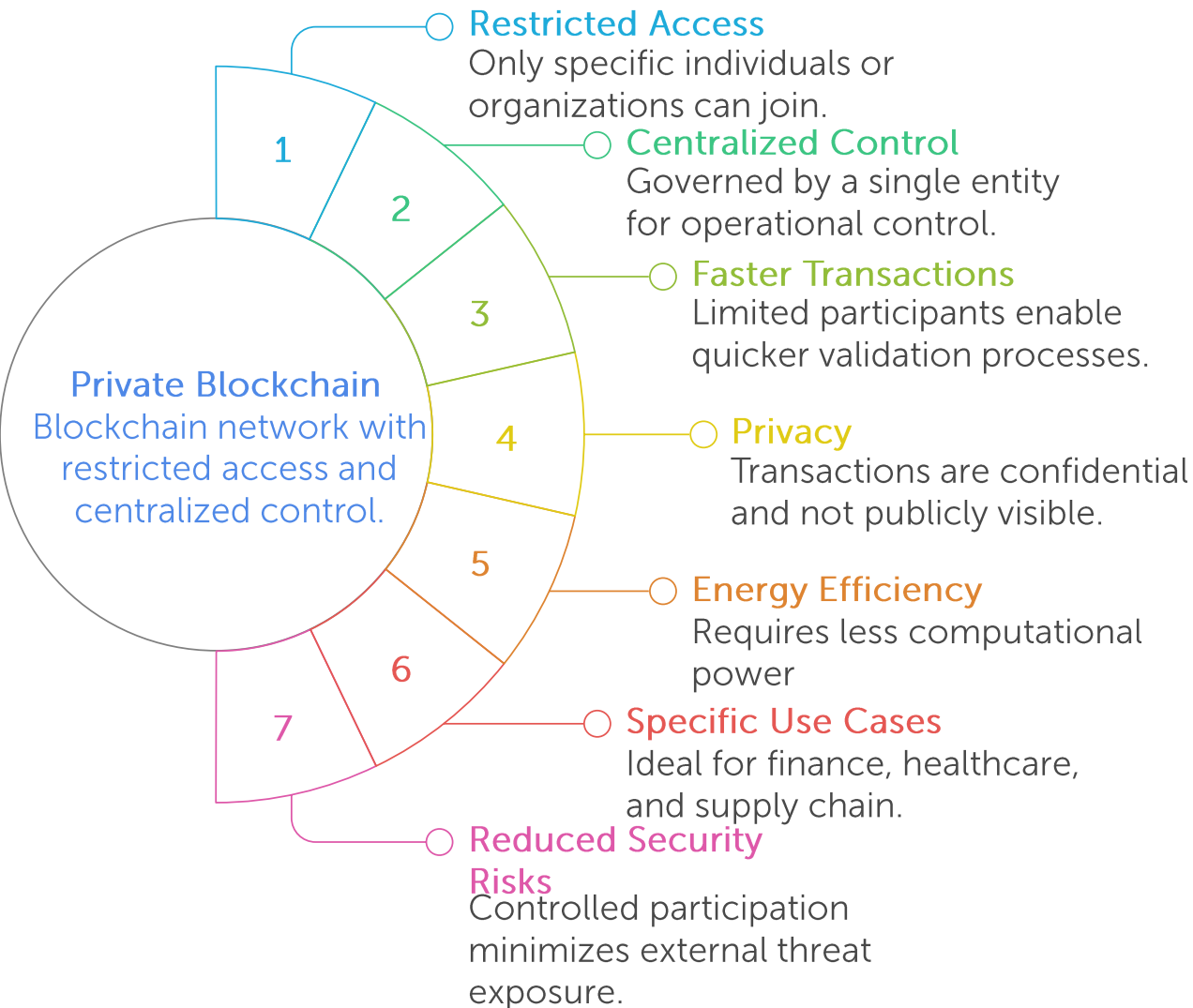
Cryptographic Security

- Blockchain uses cryptographic hashing to secure each block, ensuring that data cannot be tampered with or altered after being recorded.

How Blockchain Works: A Step-by-Step Guide



Private vs Public Blockchain



Public Blockchain



21.4 Benefits and Challenges

Challenges of Blockchain

Challenge	Description
Scalability	Struggles with speed and efficiency as networks grow, especially in Bitcoin.
Integration Challenges	Overhauling existing systems for integration is costly and time-consuming.
High Energy Consumption	Notorious for energy-intensive Proof of Work (PoW), raising environmental concerns.
Complexity	Intricate nature is a barrier, requiring technical expertise for adoption.
Interoperability	Different blockchains struggle to communicate, limiting broader applications.
Uncertain Regulations	Evolving and uncertain regulations create challenges for businesses.
Governance Issues	Decentralization leads to governance challenges, potential forks, and lack of control.
High Implementation Cost	Costly setup, development, and skilled personnel contribute to high costs.
Private Keys	Managing private keys poses security risks, especially for non-tech-savvy users.
Network Security Disruption	51% attacks compromise network integrity, especially in smaller blockchains.
Slower Process	Transaction processing is slower due to consensus mechanisms, challenging rapid environments.
Storage Problems	Expanding networks lead to significant storage demands, impacting operational costs.

Decentralization and Its Benefits

Understanding Decentralization

- Decentralization distributes control across various nodes in a network, enhancing efficiency and security.

Reducing Single Points of Failure

- By distributing control, decentralization minimizes the risk of single points of failure in the system.

Enhanced Security

- Decentralization enhances security, making data tampering extremely difficult and protecting sensitive information.

Common Vulnerabilities and Solutions

51% Attacks

- A 51% attack occurs when a single entity gains control of over half of a blockchain's mining power, compromising its security.

Smart Contract Flaws

- Smart contracts can contain programming errors that may lead to financial losses or exploitation if not audited properly.

Robust Security Measures

- Implementing robust security measures, such as regular audits and updates, is essential to safeguard blockchain integrity.

21.5 Blockchain Application and Use Cases

Financial Services and Cryptocurrency

Impact of Blockchain

Blockchain technology has transformed the financial sector by ensuring secure and transparent transactions, making it more trustworthy.

Reduction of Fraud

The use of blockchain helps in reducing fraud by providing a decentralized ledger that is nearly impossible to alter.

Rise of Cryptocurrencies

Cryptocurrencies like Bitcoin and Ethereum have gained popularity as alternative currencies, enabled by blockchain technology.



Supply Chain Management and Provenance

Blockchain in Supply Chain

Blockchain technology enhances supply chain management by providing a secure and transparent record of each product's journey.

Transparency and Trust

The use of blockchain increases transparency within the supply chain, fostering greater trust among consumers and stakeholders.

Authenticity Tracking

Blockchain allows for the effective tracking of product authenticity, ensuring that goods meet quality and safety standards.

Healthcare and Data Integrity

Ensuring Data Integrity

Blockchain technology ensures the accuracy and security of medical records, preventing unauthorized alterations and fraud.

Enhanced Data Sharing

By using blockchain, healthcare providers can share patient information securely and efficiently, facilitating better coordinated care.

Improving Patient Privacy

Blockchain enhances patient privacy through decentralized storage and controlled access, reducing the risk of data breaches.



Definition and Functionality



Smart Contracts Concept

Source: <https://www.wbs.ac.uk/news/can-smart-contracts-function-in-an-unpredictable-world/>

- ✓ Smart contracts are self-executing contracts with the agreement directly written into code, ensuring automatic execution.

Automatic Execution

- ✓ These contracts execute transactions automatically when specific conditions are met, minimizing the need for intermediaries.

Transparency and Trust

- ✓ Smart contracts enhance transparency and build trust between parties by providing a tamper-proof record of transactions.

Real-World Applications and Examples

Smart Contracts in Finance

Smart contracts are transforming the finance industry by enabling automated and secure transactions, reducing the need for intermediaries.

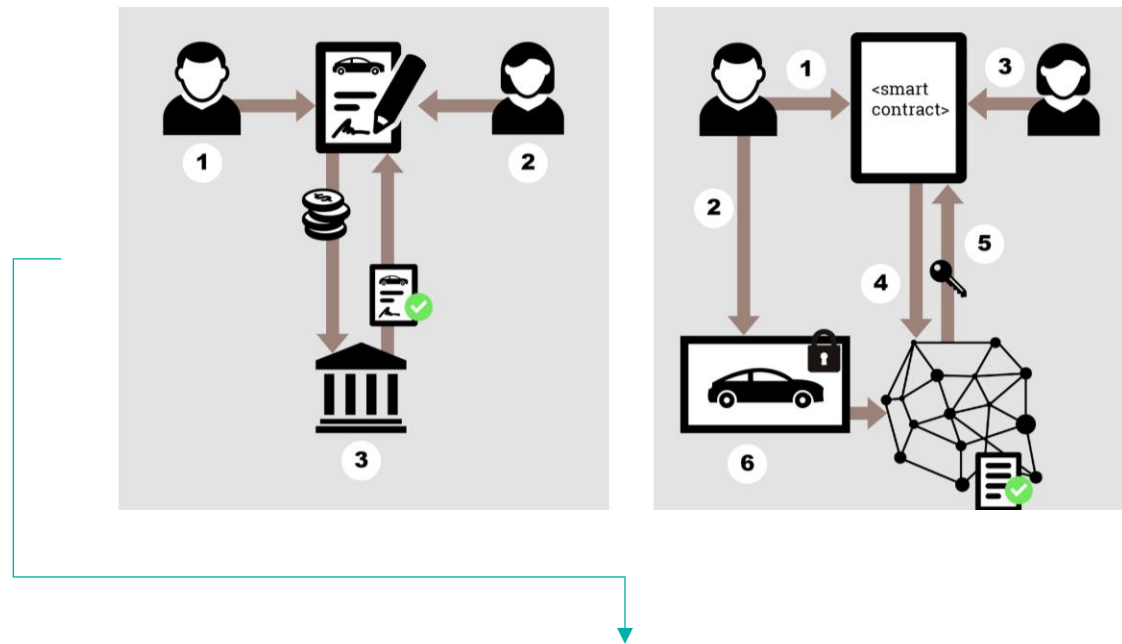
Supply Chain Management

In supply chain management, smart contracts streamline processes, increase transparency, and ensure compliance between parties.

Insurance Claims Processing

Smart contracts facilitate faster insurance claims processing by automating validation and approval, thereby enhancing customer satisfaction.

Benefits over Traditional Contracts

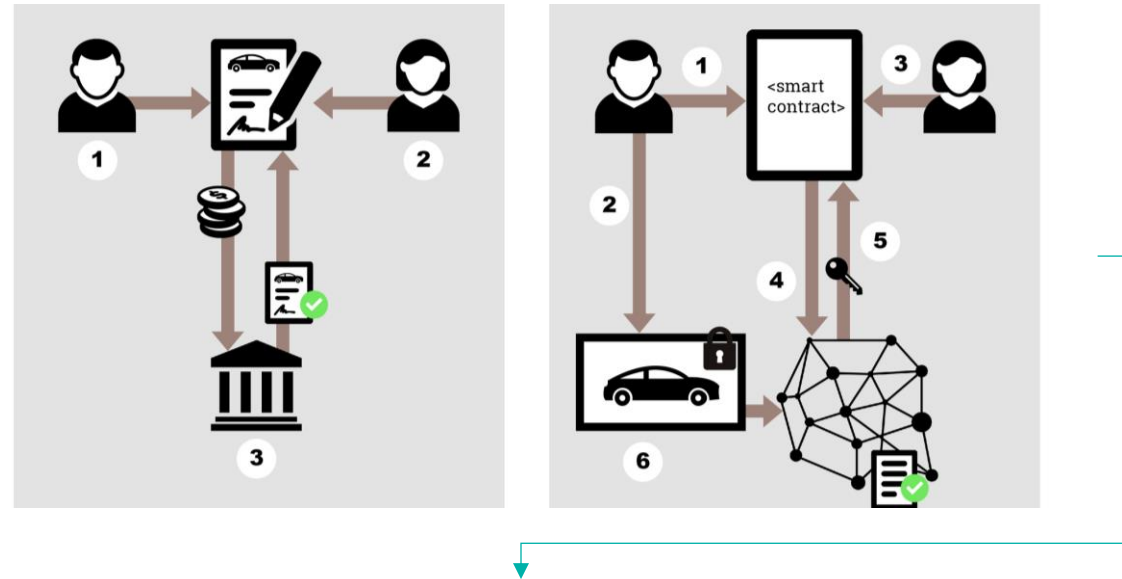


Traditional Contracts

1. Bob would like to sell his car.
2. Alice would like to buy a car.
3. A third party (intermediary) enables the trust that is needed in order to transfer the ownership of the car. Mostly different intermediaries are needed: motor vehicle registration authority, notary, insurance company. All middlemen take fees.

Source: https://link.springer.com/chapter/10.1007/978-3-031-30247-3_9

Benefits over Traditional Contracts

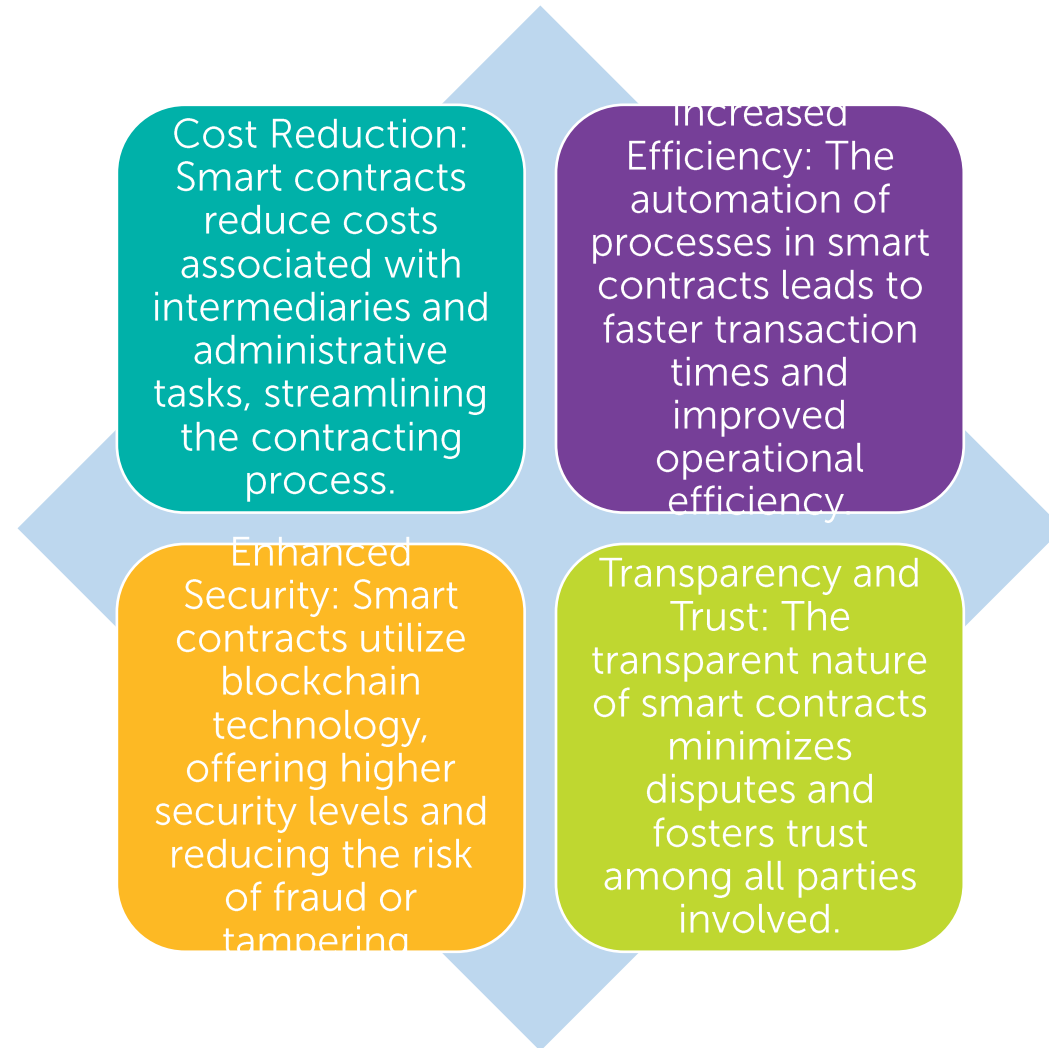


Smart Contracts

1. Bob would like to sell his car. He defines in a smart contract the conditions by which he will sell the car and signs the contract with his private key.
2. Bob leaves his car locked with a smart lock in his garage. The car has its own blockchain address and the smart lock is controlled by a smart contract.
3. Alice would like to buy a car. She finds Bob's car on an internet platform and signs the smart Bob's contract with her private key. She adds the agreed amount from her blockchain address to Bob's blockchain address.
4. As soon as the smart contract is executed the whole blockchain network will check if Bob is the real owner of the car and if Alice has enough money to buy the car.
5. If all peers in the blockchain network agree on the same state, it means that all conditions in the smart contract are met. The access code for the smart lock will be transferred to Alice and the blockchain address of the car will be registered to Alice. Bob will get the defined amount of money in his blockchain address.
6. Alice will be able to open the smart lock with her private key.

Source: https://link.springer.com/chapter/10.1007/978-3-031-30247-3_9

Benefits over Traditional Contracts



21.6 Summary

Future Developments and Standardization Efforts

Increased Standardization

Future developments in blockchain technology will focus on creating standardized practices to enhance interoperability and integration.

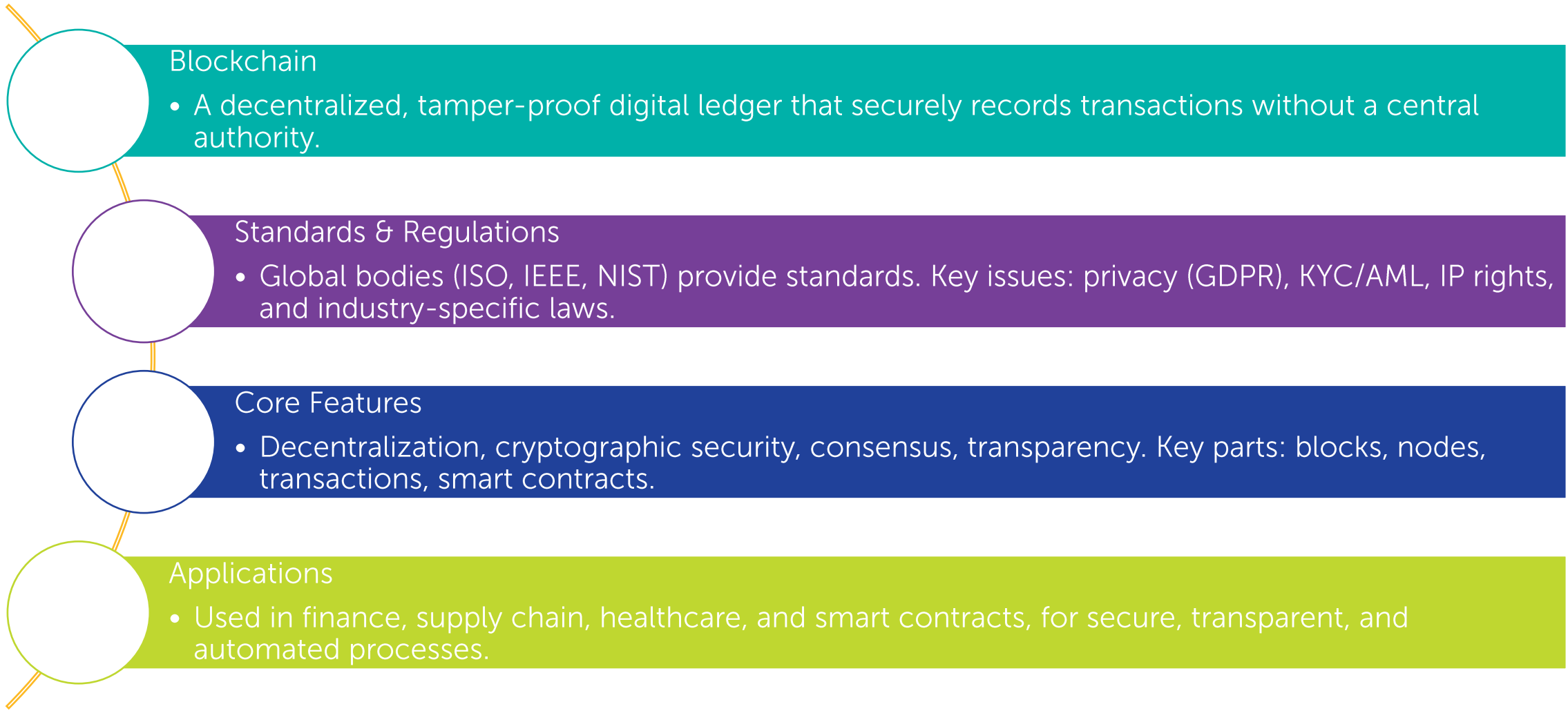
Regulatory Frameworks

Establishing regulatory frameworks is essential for blockchain's acceptance and security within various industries.

Integration into Systems

The successful integration of blockchain technology into existing systems will drive efficiency and innovation across sectors.

Summary



Post-Quiz

Interactive Learning

